

Introduction to Contract Theory and Auctions

Order Statistics

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- Let $X_1, X_2, \dots, X_n$ be n independent draws from a distribution F with associated density f , i.e., $(X_i)_{i=1, \dots, n}$ is a **random sample: i.i.d. with $X_i \sim F, i = 1, \dots, n$.**
- Let $Y_1^{(n)}, \dots, Y_2^{(n)}, \dots, Y_n^{(n)}$ be a rearrangement:

$$Y_1^{(n)} \geq Y_2^{(n)} \geq \dots \geq Y_n^{(n)}.$$

- The random variables $Y_k^{(n)}, k = 1, \dots, n$ are called the **Order Statistics**.
- Let $G_k^{(n)}$ denote the distribution of $Y_k^{(n)}$, with corresponding density $g_k^{(n)}$.
- In our study of auctions, we will frequently encounter the highest and second-highest order statistics: $Y_1^{(n)}$ and $Y_2^{(n)}$.
- Note: when the sample size n is fixed and there is no ambiguity, I will suppress the notation (n) and just write Y_k instead of $Y_k^{(n)}$, and similarly for the distribution G_k instead of $G_k^{(n)}$ and the density g_k instead of $g_k^{(n)}$.

Highest order statistic Y_1 .

- The event $Y_1 \leq y$ is the same as the event: for all k , $X_k \leq y$. Since each $X_k \sim F$ i.i.d., we have that:

$$G_1(y) = F(y)^n, \quad g_1(y) = nF(y)^{n-1}f(y)$$

$\downarrow$

$$= \prod_{k=1}^n \Pr(X_k \leq y) = F(y)^n$$

$$G_1^{(n-1)}(y) = F(y)^{n-1}$$

$$g_1^{(n-1)}(y) = (n-1) F(y)^{n-2} f(y)$$

Second-highest order statistic Y_2 .

- The event that $Y_2 \leq y$ is the union of the following disjoint events:

$$\{X_k \leq y, \forall k = 1, \dots, n\}$$

OR

$$\{n - 1 \text{ of the } X_k \text{ are less than or equal to } y \text{ and one is greater than } y.\}$$

- There are $\binom{n}{n-1} = n$ ways that the second event can happen, thus:

$$\begin{aligned} G_2(y) &= F(y)^n + nF(y)^{n-1}(1 - F(y)) \\ &= nF(y)^{n-1} - (n - 1)F(y)^n \end{aligned}$$

- The density function is:

$$g_2(y) = n(n - 1)(1 - F(y))F(y)^{n-2}f(y)$$

Relationships between the highest and second-highest order statistics.

Note that

$$\begin{aligned} G_2^{(n)}(y) &= nF(y)^{n-1} - (n-1)F(y)^n \\ &= nG_1^{(n-1)}(y) - (n-1)G_1^{(n)}(y) \end{aligned}$$

and thus, the density satisfies:

$$g_2^{(n)}(y) = ng_1^{(n-1)}(y) - (n-1)g_1^{(n)}(y)$$

The above immediately implies that

$$\mathbb{E}[Y_2^{(n)}] = n\mathbb{E}[Y_1^{(n-1)}] - (n-1)\mathbb{E}[Y_1^{(n)}]$$

We also have that:

$$\begin{aligned} g_2^{(n)}(y) &= n(n-1)(1-F(y))F(y)^{n-2}f(y) \\ &= n(1-F(y))g_1^{(n-1)}(y) \end{aligned}$$

Joint and Conditional Distributions of Order Statistics

- Recall that although $X_1, \dots, X_n$ are independent, the order statistics are not.
- Let $\mathbf{Y} = (Y_1^{(n)}, \dots, Y_n^{(n)})$ be the vector of order statistics.
- Then the **joint density** of $\mathbf{Y}$ is

$$g_{\mathbf{Y}}(y_1, \dots, y_n) = \begin{cases} n! f(y_1) f(y_2) \cdots f(y_n) & \text{for } y_1 \geq y_2 \geq \cdots \geq y_n \\ 0 & \text{otherwise.} \end{cases}$$

$x_1, x_2, x_3, [0,1]$

- From here we deduce¹ that the joint density of the first- and second-order statistics is

HW $n=3$
 $g(y_1, y_2, y_3) = 6 f(y_1) f(y_2) f(y_3)$
 $y_1 \geq y_2 \geq y_3$
 $\hookrightarrow g_{1,2}^{(3)}(y_1, y_2)$

$$g_{1,2}^{(n)}(y_1, y_2) = \begin{cases} n(n-1) f(y_1) f(y_2) F(y_2)^{n-2} & y_1 \geq y_2 \\ 0 & \text{otherwise} \end{cases}$$

¹This is done by integrating out all the other variables using the appropriate ranges of integration. You will need to use integration by parts many times.

Conditional distributions

- The density of $Y_2^{(n)}$ conditional on $Y_1^{(n)} = y$ is

$$\begin{aligned} g_2^{(n)}(z|Y_1^{(n)} = y) &= \frac{g_{1,2}^{(n)}(y, z)}{g_1^{(n)}(y)} \\ &= \frac{n(n-1)f(y)f(z)F(z)^{n-2}}{nF(y)^{n-1}f(y)} \\ &= \frac{(n-1)f(z)F(z)^{n-2}}{F(y)^{n-1}} \end{aligned}$$

if $y \geq z$ and 0 otherwise.

- The density of $Y_1^{(n-1)}$ conditional on the event $Y_1^{(n-1)} < y$ is

$$g_1^{(n-1)}(z|Y_1^{(n-1)} < y) = \frac{g_1^{(n-1)}(z)}{G_1^{(n-1)}(y)} = \frac{(n-1)f(z)F(z)^{n-2}}{F(y)^{n-1}}$$

- Thus $g_2^{(n)}(z|Y_1^{(n)} = y) = g_1^{(n-1)}(z|Y_1^{(n-1)} < y)$ for all $z < y$